

Muhammad Abdullah

Structural Engineer

0336-4771103 | abdullah8t90@gmail.com | [Portfolio \(PDF\)](#)

EDUCATION

University of Illinois Urbana-Champaign

MS Civil Engineering (Structural)

Coursework: Advanced Steel Design, Structural Dynamics, Finite Element Analysis

Aug. 2024 – May 2026

CGPA: 3.93

Military College of Engineering, NUST

BE Civil Engineering

Oct. 2017 – Jul. 2021

CGPA: 3.82

EXPERIENCE

Graduate Researcher

Smart Structures Technology Lab, University of Illinois

Mar. 2025 – May 2026

Urbana, IL, US

- Integrated reliability constraints into a legacy stochastic topology optimization framework.
- Implemented and validated adjoint sensitivity formulations to compute gradients for highly non-linear reliability constraints.
- Refactored core MATLAB scripts into a modular architecture to allow for greater flexibility in adding components to the framework.

Management Trainee Engineer

National Logistics Cell

Feb. 2022 – Mar. 2023

D.I. Khan

- Managed execution of structural and road components in highway projects and inspection of bridge elements
- Responsible for supervising subcontractors, and coordinating between consultant and contractor staff.
- Prepared daily work reports, presentations, and updated monthly work schedules and progress.
- Prepared bar bending schedules for building construction projects, and coordinated procurement.
- Prepared a hydraulic report proposing an alternate design of a large culvert which was accepted. Alternate design resulted in significant time and cost savings for the contractor.
- Created an Android app linked to Google Sheets to track rebar inventory and issue details for multiple subcontractors, providing real-time summaries of currently held and issued rebar by type.

Intern

Designmen Consulting

Nov. 2021 – Jan. 2022

Islamabad

- Gained familiarity with high-rise residential tower drawings/plans and assisted with inspection of site-work.

PROJECTS

Optimization of Viscous Damper Distribution in a 3D Frame Structure

- Developed a MATLAB-based topology optimization framework to determine the optimal distribution of viscous dampers in a 3D asymmetric frame structure under lateral drift and rotation objectives.
- Extracted and processed structural matrices from SAP2000 to rebuild the model inside MATLAB, creating a more effective pipeline while testing different variations in the base structure.

AWARDS

Fulbright Foreign Student Scholarship

U.S. Department of State & USEFP

Aug. 2024 – May 2026

- Received a fully funded scholarship for an MS program in Civil Engineering at the University of Illinois Urbana-Champaign.

TECHNICAL SKILLS

Software: ETABS, SAP2000, Tekla Structures, STAAD.Pro, Excel, Revit, AutoCAD, Primavera, QGIS, HEC-RAS

Computational: Python, MATLAB, Grasshopper